Fiscal Year:	FY 2004	**Task Last Updated:**	FY 10/24/2006
PI Name:	Linnell, Bruce		
Project Title:	Advanced Fire Detection In Enclosed Environment Via The Detection Of Chemical Vapor Signatures From Pre-Combustion Processes Using Machine Olfaction		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bruce.linnell-1@ksc.nasa.gov	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	
Organization Name:	ASRC Aerospace		
PI Address 1:	Chemical Instrumentation and Processes		
PI Address 2:	ASRC-15		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899-0001	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-B
Start Date:	10/01/2003	**End Date:**	09/30/2005
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: See Immer, Christopher for FY2006 final report (info per GRC 12/07); end date for this project changed to 9/30/2005 from 9/30/2006 (11/2010). NOTE: Added to Task Book in 10/2006 per GRC		
Key Personnel Changes/Previous PI:	NOTE: SEE IMMER, CHRISTOPHER. PI changed to Christopher Immer from Bruce Linnell, sometime during FY2006, per GRC contact information (12/07).		
COI Name (Institution):	Stetter, Joseph R (Illinois Institute of Technology) Hong, Todd C (NASA JSC) Young, Rebecca (NASA KSC)		
Grant/Contract No.:	NNC04CB08C		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The detection of fires is very important within closed environments such as the Space Shuttle and International Space Station. Fires need to be detected within such areas as soon as possible, preferably before combustion has taken place. Previous research performed at KSC has shown that the multi-sensor technology commonly known as the electronic nose (e-nose) is able to detect the offgassing products from the insulation of heated wires well before combustion starts. Note that e-nose technology is not concerned with identifying in an analytical sense the chemicals being detected, but with classifying a mixture of vapors as coming from one source or another. This project would expand upon these results, and produce an integrated system of sensors and software which would not only provide a pre-combustion alarm, but also identify what kind of material is being overheated (wire insulation, printed circuit board, etc.). Metal oxide semiconductor (MOS) sensors are low cost, available in numerous miniature configurations, have a wide range of selectivity to various vapors, and are simple to operate. To that end, commercially available MOS sensors will be incorporated into a sensor array and exposed to thermal out-gassing products of electrical components. In addition, conventional circuit board components (e.g., IC chips-usually encapsulated in resins) will be subjected to heat cycles to mimic catastrophic failure with a potential fire hazard. Pattern recognition techniques will be developed at KSC to identify those commercially available sensors which provide the most sensitivity and reliability to detection and identification. It is recognized that other sensor technologies (e.g., SAW, quartz micro-balance, etc.) could potentially increase the identification power of the array, and so other sensor types will be tested and incorporated into the array if they are found to be useful. While this technology would be designed with flight consideration in mind, the result of this project will not be a flight-ready system, but one or two prototypes whose sensors have been partially screened for flight qualification.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fire detection.
Task Progress:	New project for FY2004; no report received during this reporting period. New to Task Book in 2006. PI changed at end of FY 2005; see Immer, Christopher, for final report information in FY2006 records.
Bibliography Type:	Description: (Last Updated:)